

C1 · Parsing-Retrieval Disconnect

Candidate parsers



Qwen3-VL



MinerU



Marker



PaddleOCR

Cleaner-looking $\neq$ better retrieval

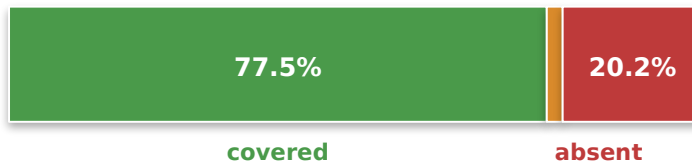
parser	Boundary Clarity	answer	Hit@1
MinerU	0.72	no	0.20
Prod (ours)	0.61	yes	0.55

parser choice alone swings Hit@1 by **2.8×**
appearance metrics measure formatting, not content
(anti-correlation $r = -0.81$, $n = 5$)

C2 / C3 · Diagnose + Select

C2 · Coverage diagnostic (no retriever)

where does each gold answer land?



absent = answer never made it into the parser output
(parser fault)
constant across all 8 chunkers $\rightarrow$ chunking can't recover it

C3 · RCPS selection protocol (no training)

held-out Q-A probe, scored over 3 retrievers:



BGE-M3



mE5



Qwen3-Emb

$\rightarrow$ rank parsers & chunkers by retrieval

C4 · Bounded Parser Training

C4 · Parser-side training

best-of-K parses from Prod, ranked two ways:

RADP-Distill — edit-distance $\rightarrow$ GT

RADP-DPO — page-local RCPS

Result

Hit@5 gain over the untuned parser:

RADP-Distill  **+1.22 pp**

RADP-DPO  **+0.85 pp**

$\approx$ **overlapping CIs $\rightarrow$ equal gain**
retrieval reward unnecessary; lever = fidelity distillation